Three-dimensional data of wire-cut surface scans under the confocal microscope

Yuhang Lin[1,2,*], Heike Hofmann[2,3], Curtis Mosher[4], Eden Amin[5], Jeff Salyards[2], Alicia Carriquiry[1,2]

[1]Department of Statistics, Iowa State University, USA
[2]Center for Statistics and Applications in Forensic Evidence, Iowa State University, USA
[3]Department of Statistics, University of Nebraska-Lincoln, USA
[4]Roy J. Carver High Resolution Microscopy Facility, Iowa State University, USA
[5]Forensic Science Institute, University of Central Oklahoma, USA
[*]Corresponding author: Yuhang Lin (yhlin@iastate.edu)

Abstract

Wire-cut striation marks can provide evidence for forensic source identification, but publicly available datasets and reproducible quantitative workflows for analyzing these marks are limited. We present a dataset of 120 three-dimensional topographic scans in `x3p` format from aluminum wires cut with five KWS-105 wire cutters. The study contains six scans—two replicates at each of three cutting locations—for each of the 20 tool-edge combinations. We provide additional scan metadata, manually extracted profiles, processed signals, cross-correlation function (CCF) values for pairwise aligned signals, images, and analysis code. The accompanying workflow covers scan inspection, profile and signal extraction, alignment, pairwise similarity visualization, CCF distributions for same-source and different-source comparisons, and variability assessment across replicates, stagings, and acquisitions. The dataset and workflow support the development, evaluation, and reproducible comparison of quantitative methods for wire-cut evidence.

Background & Summary

In forensic analysis, marks left at crime scenes play an important role, and forensic examiners are particularly interested in identifying their origin, a process known as the Source Identification Problem in Forensic Science[1]. The forensic science community focuses on two distinct problems: the **specific source problem**, in which examiners aim to determine whether a mark was left by a particular tool, and the **common source problem**, in which examiners aim to determine whether two marks were left by the same tool. Currently, accepted practice for both problems relies on visual inspection of physical items under a comparison microscope. Examiners then report their conclusions according to the Theory of Identification[2], developed by the Association of Firearm and Toolmark Examiners (AFTE), using one of three categories: *identification*, in which the marks are believed to have been made by the same source; *elimination*, in which the marks are believed to have been made by different tools; and *inconclusive*, in which the similarities or dissimilarities between marks are insufficient to support either identification or elimination. This process has been criticized by the National Research Council (NRC)[3] and the President's Council of Advisors on Science and Technology (PCAST)[4] for its lack of objectivity and absence of quantifiable measurements, such as error rates.

In discussing error rates, Biedermann[5] differentiates between internal and external perspectives. The external perspective allows general statements based on black-box studies that relate examiners' conclusions to ground truth; however, it does not consider the evidence in a particular case. Introducing an internal perspective requires evaluating similarity between pairs of specimens with known ground truth. Algorithms allow these similarities to be quantified objectively and reproducibly.

Advances in microscopy have made it possible to move from optical two-dimensional imaging to digital three-dimensional topographic scanning, allowing algorithmic assessment of the

Figure 1. Microscopic wire-cut striations. Close-up of striations left by a blade on the cut end of a wire.

similarity between pairs of forensic pattern evidence[6]. This transition has already supported practical advances in bullet and cartridge comparisons, including visual inspection of bullet line scans from three-dimensional topographic images[7], the use of features extracted by fast Fourier transform (FFT) as the basis of an artificial intelligence system[8], an automated pilot study based on consecutively matching striae[9] (a feature also used by some laboratories in casework), and the incorporation of three-dimensional imaging into routine casework at the Alabama Department of Forensic Sciences (ADFS)[10].

When cut wires are found at a crime scene, understanding the characteristics of marks left by bladed tools becomes essential. When a blade cuts a wire, it leaves striations on the surface, as shown in Figure 1. These striations provide a basis for similarity assessment and source identification. Similar scans have supported the development of quantitative methods for analyzing firearms and other toolmark evidence. For example, studies of screwdriver striations have examined key factors affecting comparisons, including angles of attack[11], rotational axes[12], and cutting directions[13]. Foundational research on forensic toolmarks has also emphasized data collection and analysis, including the collection and distribution of datasets for bullet and toolmark analysis[14,15], numerical methods for improving accuracy and consistency[16,17], and techniques for quantifying similarity[18,19].

Data collected at crime scenes lack the known ground truth needed to estimate error rates; therefore, controlled studies are needed. In this study, we provide a publicly available dataset of wire-cut scans together with a systematic quantitative processing and comparison workflow. The shared materials include raw scans, metadata, manually extracted profiles, processed signals, pairwise cross-correlation function (CCF) values, derived images, and analysis code.

The Methods section describes sample collection, scanning, profile extraction, signal processing, and CCF-based alignment. Data Records describes the shared files and repository organization. Data Overview presents descriptive summaries of pairwise CCF values. Technical Validation assesses signal-processing and alignment behavior and variability across replicates, stagings, and acquisitions. Usage Notes provides practical guidance for working with the shared files, and Code Availability identifies the custom processing and validation code. The dataset and workflow may support future quantitative research on wire-cut evidence.

Methods

The cutting tools were five Kaiweets KWS-105 wire cutters. The material was general-purpose, 16-gauge, 1.5-mm anodized pure aluminum wire sold for crafting. **[Agent: Provide the wire manufacturer, product name, or another source identifier if available so readers can identify the material used.]** The physical properties of aluminum wire make it suitable for retaining marks while remaining relatively easy to bend and nontoxic. **[Agent: Confirm or qualify the statement that the wire was nontoxic; if this is a comparative safety claim rather than a measured material property, revise it and cite an appropriate source.]** The aluminum wire has a hardness of 15 HB on the Brinell Hardness scale. This hardness allows the tools to leave marks while helping the cut samples resist damage during handling under the microscope. For comparison, the hardness values of pure lead and copper are 5 HB and 35 HB, respectively. **[Agent: Confirm whether 15 HB was measured for the wire used. If it is a literature value for pure aluminum rather than this anodized wire, revise the statement accordingly and cite the source for all three hardness values.]** In casework, aluminum wire is less common than lead or copper wire. **[Agent: Add a supporting source for the relative frequency of aluminum, lead, and copper wire in casework, or qualify the statement if it is based on informal experience. If retained, this contextual claim belongs in Background & Summary rather than Methods.]** We avoided using lead in the laboratory because of its toxicity.

Cutting wires

We used 16-gauge, 1.5-mm anodized aluminum wire to create this wire-cut dataset. For each of the five wire cutters, we made two replicate cuts at each of three blade locations (inner, middle, and outer). **[Agent: State how the inner, middle, and outer cutting locations were selected or marked, if this information was recorded.]** Figure 2 (a) shows the three blade locations. A 4-inch wire segment was used for each replicate and divided into two 2-inch sides, AB and CD. Each side was separated along the bend into two pieces. The resulting pieces comprise four cut surfaces (A, B, C, D). As shown in Figure 2 (c), C is the reverse of A and D is the reverse of B. The complete design therefore yields 5 tools $\times$ 3 locations $\times$ 2 replicates $\times$ 4 cut surfaces = 120 scans. **We scanned each cut surface using a [manufacturer and model] confocal microscope with [objective or magnification]. Each scan covered [scan area or pixel dimensions] at a lateral sampling interval of 0.645 micrometers per pixel and a vertical resolution of [value].** **[Agent: Replace the bracketed microscope, magnification, scan-size, and vertical-resolution placeholders.]** The scans were saved in `x3p` format. The microscope is shown in Figure 2 (b). Each piece was labeled according to the following naming convention:

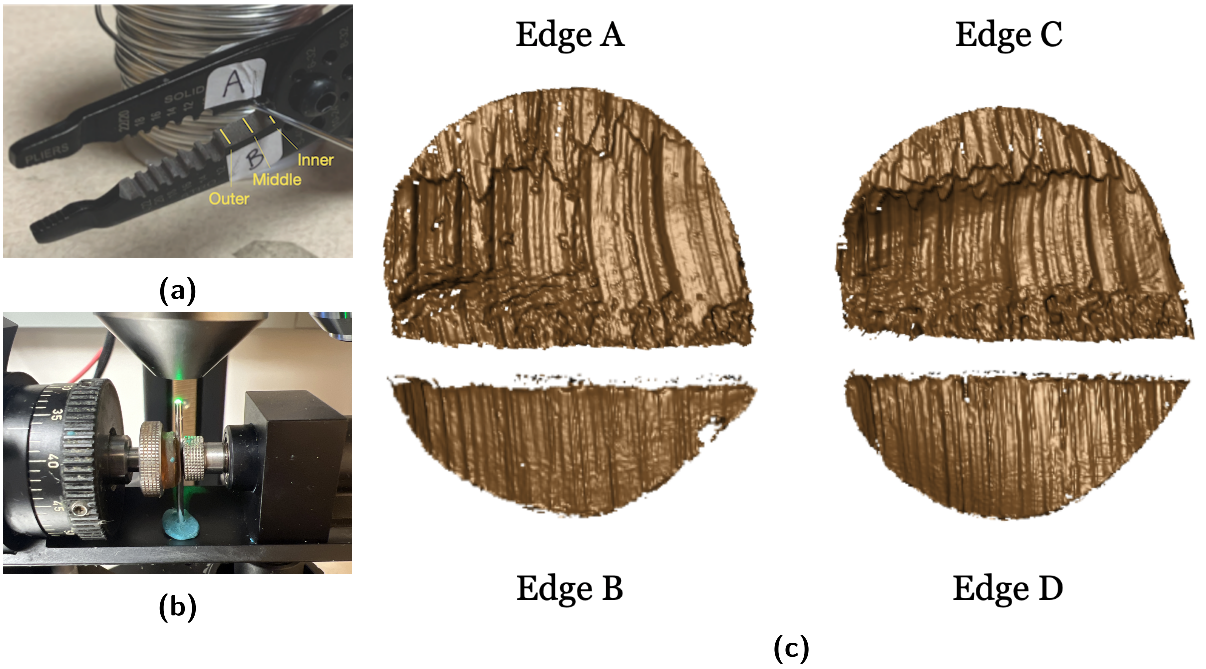

Figure 2. Wire-cut sampling and scanning workflow. (a) A Kaiweets KWS-105 wire cutter with inner, middle, and outer locations marked. (b) A confocal microscope used to scan wire-cut surfaces. (c) Four cut-surface samples labeled A, B, C, and D after separation.

```
T__W-L_-R_
 ||   |  |
 ||   |  +-- Replicate: 1 or 2
 ||   +----- Location: I (inner), M (middle), or O (outer)
 |+--------- Edge: A, B, C, or D
 +---------- Tool: 1, 2, 3, 4, or 5
```

The filename components are T(ool) 1/2/3/4/5, (Edge) A/B/C/D, W(ire), L(ocation) I(nner)/M(iddle)/O(uter), and R(eplicate) 1/2. For example, T1AW-LI-R1 denotes the first replicate of a piece cut by tool 1 on edge A at the inner location. To minimize errors in the sampling protocol, two team members cut and labeled the wires together. One team member then scanned the samples in a specified order and named the scans according to the naming convention, and a third team member checked the saved data.

Extract profiles

Numerical comparisons between two replicates cannot be performed directly on the `x3p` files. We first extracted representative profiles from the scans. A profile is a sequence of surface-height values along a user-drawn line. After processing, each profile became a signal representing one scan and was used for subsequent comparisons. Using `x3ptools::x3p_extract_profile`, we manually drew one line across each scan, perpendicular to the dominant striation direction, and exported the surface-height values and extraction coordinates. An example profile line is shown in dark blue in Figure 3 (a), and the resulting profile function is plotted in Figure 3 (b).

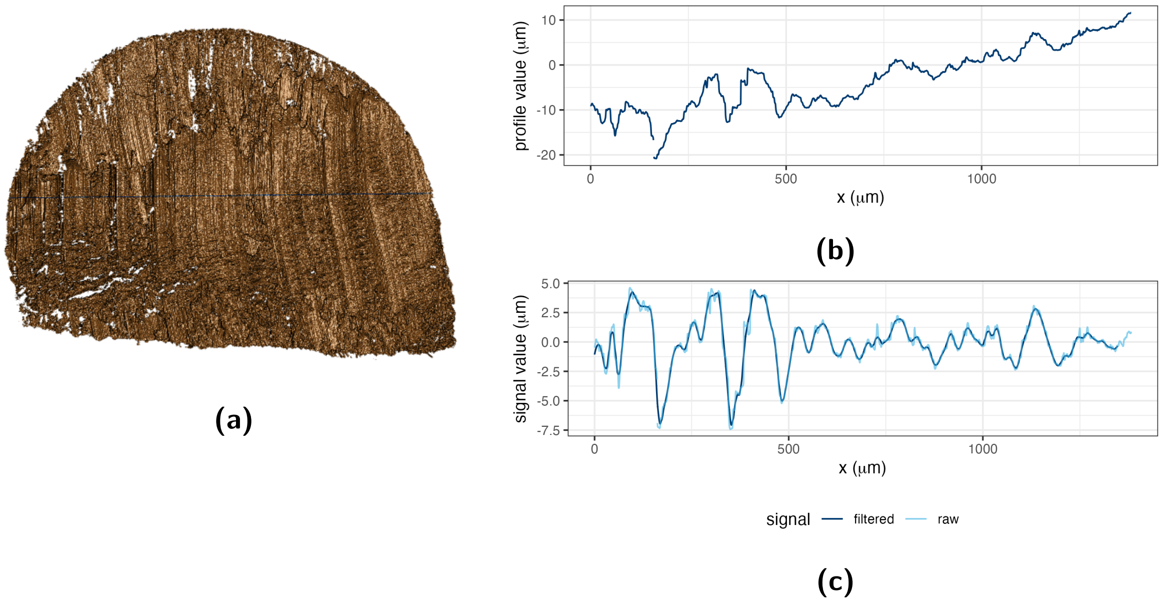

Figure 3. Profile and signal extraction. (a) A dark-blue profile line drawn across the scan striations. (b) The profile function extracted along the line in (a). (c) Raw and filtered signals obtained from the profile function in (b).

Filtered signals

After extracting each profile, we applied a two-span smoothing procedure, with span parameters of 400 and 40 controlling removal of the broad trend and smoothing of local variation, respectively[20]. We then checked for extreme deviations at the ends of the signal. The horizontal coordinates were divided at the 0%, 2.5%, 5%, 95%, 97.5%, and 100% quantiles. At each end, a two-sample t-test compared the outermost 2.5% interval with the adjacent interval. When the boundary two-sample t-test produced a p-value below 0.01, values in the outermost interval were removed. The resulting smoothed signal was used for subsequent comparisons. Figure 3 (c) shows the resulting smoothed signal.

Align signals

After extracting signals from different scans, we compared all possible signal pairs. For each pair, we calculated the CCF across relative shifts and retained the shift with the maximum CCF as the optimal alignment. The comparison between T1AW-LI-R1 and T1AW-LI-R2, for example, corresponds to a row in Figure 4. The first two columns of Figure 5 show the extracted signals, and the rightmost column shows their alignment and CCF value.

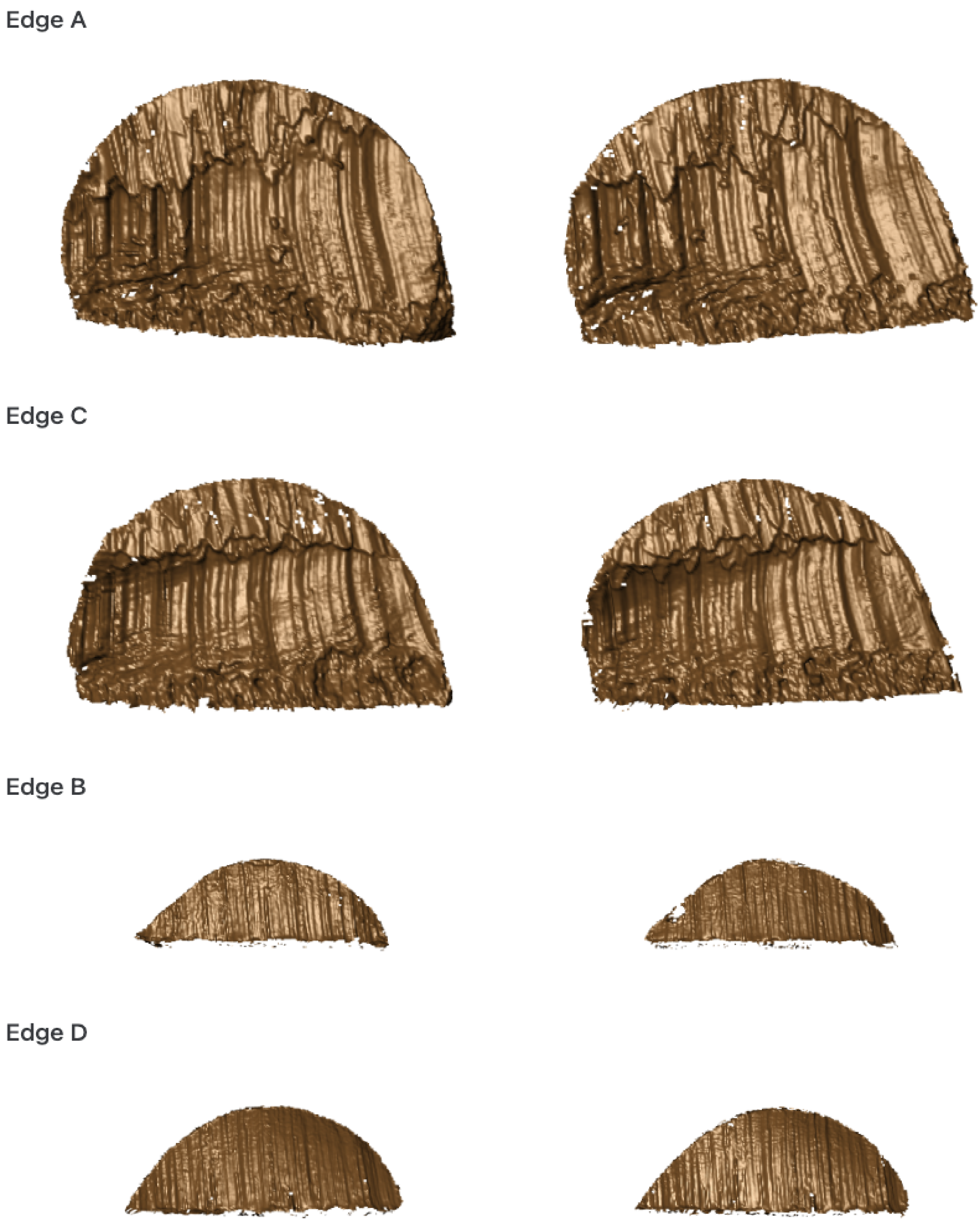

Figure 4. Same-source replicate scan pairs. Each row shows the two replicate scans for one cut surface (A, C, B, or D) from tool 1 at the inner blade location.

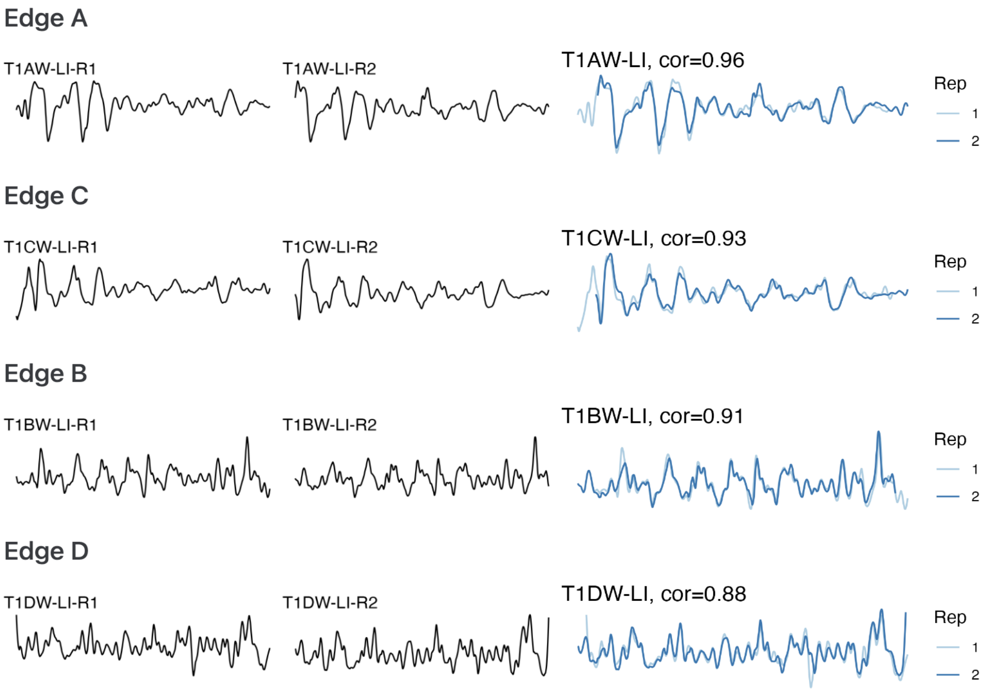

Figure 5. Same-source signal alignment. Rows correspond to cut surfaces A, C, B, and D. The first two columns show signals extracted from the replicate scans in Figure 4; the third column overlays the aligned signals and reports their CCF values.

Data Records

The complete dataset, including the `x3p` scans, metadata, extracted profiles, processed signals, pairwise CCF values, and derived images, is available from Iowa State University DataShare [DATASET CITATION] at `https://doi.org/[DOI]`. [Agent: Replace the dataset citation and DOI placeholders. For first-round review, the destination must permit anonymous download; from the second round onward, it must be a public formal repository record. Add the full dataset reference to References.]

The repository contents, formats, and folder organization are summarized in Table 1.

A data dictionary defining non-obvious columns in `meta.csv`, the profile and signal files, and the pairwise-CCF file is provided in [repository documentation path]. The dataset is available under the [license] license. [Agent: Verify Table 1 against the deposited record and replace the documentation-path and data-license placeholders. For this non-sensitive original dataset, use CC0 or CC BY rather than a license containing NC or SA restrictions.]

Table 1. Structure of available data and files.

Data type	Description	Section
Raw data		
scans/	folder containing 120 topographic 3D scans corresponding to 30 aluminum wire cuts (exttt{x3p} format)	Cutting wires
meta.csv	metadata for each cut, including tool, blade, and location information (CSV format)	Cutting wires
Manual derivatives		
profiles/	folder containing manually extracted profiles (CSV format)	Extract profiles
Computational derivatives in the folder 'data-derived/'		
wire-signals	signals processed from the corresponding profiles (zipped CSV format)	Filtered signals
wire_pairwise_ccf	CCF values for all pairwise aligned signals (zipped CSV format)	Align signals
Image files		
pngs/	folder containing images of 3D wire-cut scans (PNG format)	Cutting wires
profile-images/	folder containing images of profiles extracted from wire cuts (PNG format)	Extract profiles
Visual inventory in the folder 'assessment/analysis-manual/'		
processing-wires	display of pairwise aligned signals from the same sources (HTML format)	Align signals
Technical validation		
variability-assessment/	folder containing technical-validation documents, with an internal structure that mirrors the main folder	Technical Validation

Data Overview

As descriptive summaries of comparison behavior across the dataset, Figure 6 displays pairwise CCF values organized by tool, cut surface, blade location, and replicate. The main diagonal contains self-comparisons, whereas the off-diagonal cells compare distinct signals. Figure 7 displays the CCF distributions for same-source and different-source comparisons.

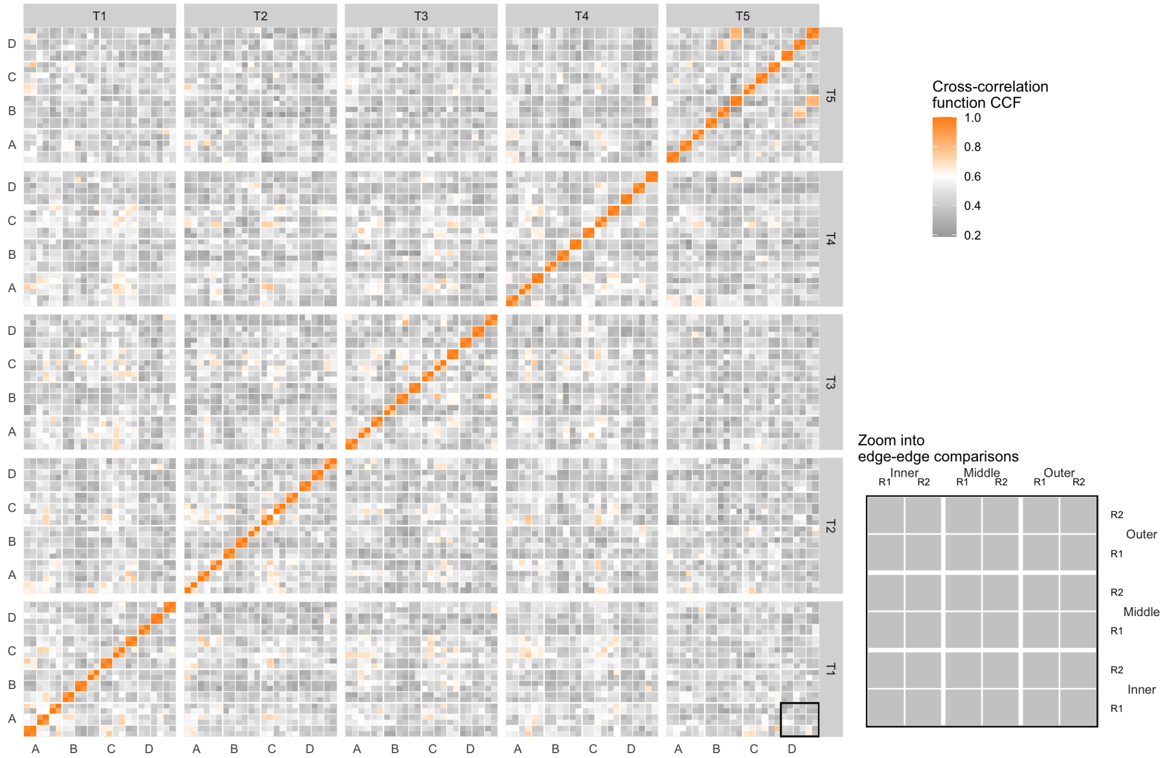

Figure 6. Pairwise CCF tile map. Rows and columns group scans by tool, cut surface, blade location, and replicate. Orange cells indicate larger CCF values and gray cells indicate smaller values; the inset enlarges the boxed edge-to-edge comparison block.

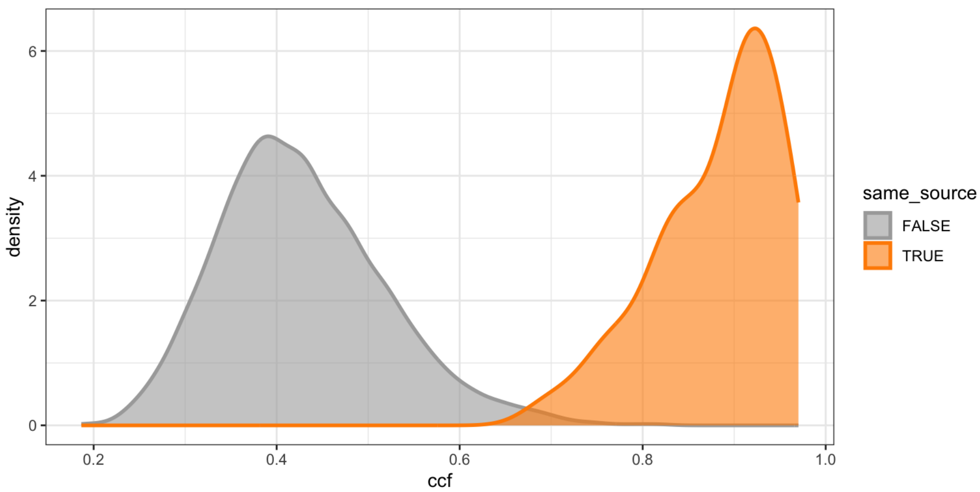

Figure 7. CCF density distributions. Kernel density estimates are shown for different-source comparisons in gray and same-source comparisons in orange.

Technical Validation

We conducted technical validation in two areas: (1) signal processing and alignment and (2) variability assessment.

We validated scan processing and signal alignment as follows. We expected high CCF values between signals from the same source and low values between signals from different sources. Figure 5, shown earlier in Align signals, meets our expectation for same-source signals. As an example of a different-source comparison, we selected T1AW-LI-R1 and T2AW-LI-R1. Figure 8 (a) and (b) show the selected scans. Their signal alignment, shown in Figure 8 (c), has a CCF value of 0.2, which is as low as expected. For the following summaries, edges A and C were categorized as long, edges B and D as short, and comparisons pairing one long edge with one short edge as mixed. We also summarized the resulting CCFs for all pairwise comparisons using boxplots and receiver operating characteristic (ROC) curves. Figure 9 and Figure 10 show these summaries. The ROC curves plot sensitivity, or the true positive rate, against the false positive rate (FPR; 1 - specificity). The curves are close to the upper-left corner, indicating strong classification performance. For the combined comparison set, a threshold of 0.589 controls the FPR below 0.05 with a false negative rate (FNR) of 0, whereas a threshold of 0.658 controls the FPR below 0.01 with an FNR of 0.02. These results are consistent with our expectations and validate our pipeline for processing and aligning signals.

The second validation assessed variability in the pipeline across replicates, stagings, and acquisitions. We selected scans from tool 2 at the middle location for this assessment. Replicate variability was assessed using two different cuts at the same edge and position. The comparison in Figure 5 in Align signals illustrates this assessment. To assess staging variability, we rescanned replicate 2 under the confocal microscope and introduced the staging label S into the naming protocol. The S1 designation is omitted from labels, so R2 is equivalent to R2-S1. For all four edges of tool 2 at the middle location, we created three stagings (R2, R2-S2, and R2-S3), computed pairwise CCFs between stagings, and calculated their mean and standard deviation (SD).

Within each staging, we further examined the effect of different acquisitions by keeping the samples on the confocal microscope while varying the lighting conditions. We introduced A as the acquisition label. For edge A at staging 3, we performed four acquisitions and omitted the A1 designation from the first acquisition label. Therefore, R2-S3 is equivalent to R2-S3-A1. For the other three edges, we conducted two acquisitions at staging 3. These acquisitions are denoted by R2-S3 and R2-S3-A2. Within staging 3, we computed pairwise CCFs between acquisitions and calculated their mean and standard deviation. Variability between distinct stagings was assessed separately through the staging comparisons described above.

With this setup, we expected CCF values to become more consistent from replicate to staging to acquisition as the scanning conditions became progressively more stable. Consequently, variability was expected to decrease along this sequence. We also compared replicates, stagings, and acquisitions, expecting smaller differences between acquisitions and stagings than between stagings and replicates. We visualized the results as scatter plots. Figure 11 shows the scatter plots and intervals extending two standard deviations above and below the mean. Higher orange mean lines indicate an increase in the mean CCF, and narrower shaded intervals indicate a decrease in the standard deviation. The detailed numerical results, including means, standard deviations, and differences between means, are shown in Table 2 and confirm our expectations.

[Agent: The manuscript contains 11 figures. Scientific Data recommends no more than eight but does not mandate this limit. Confirm with the coauthors or professor whether any logically connected pairs should be combined for revised submission; do not remove a figure or its discussion without an author decision.]

Our variability assessment does not include variation in CCF values caused by differences in the manual extraction of profiles from scans. To support reproduction of the results reported

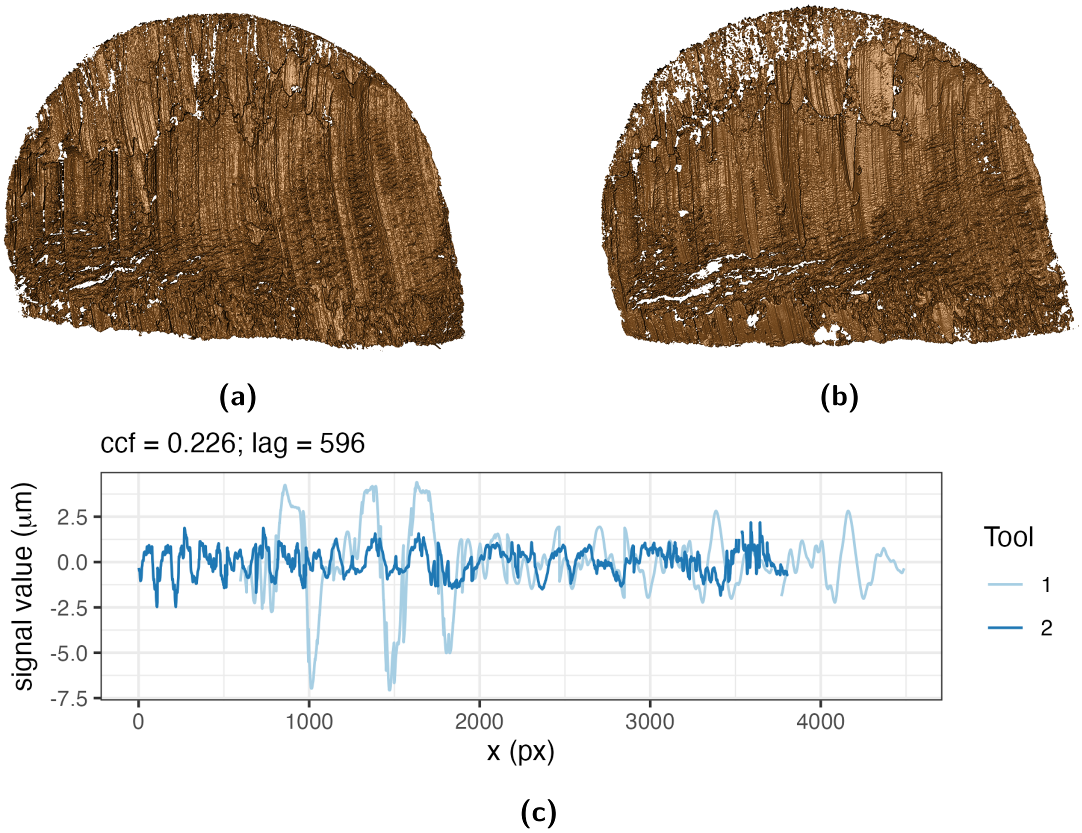

Figure 8. Different-source scan and signal comparison. (a) Scan T1AW-LI-R1 cut by tool 1. (b) Scan T2AW-LI-R1 cut by tool 2. (c) Alignment of signals from T1AW-LI-R1 and T2AW-LI-R1.

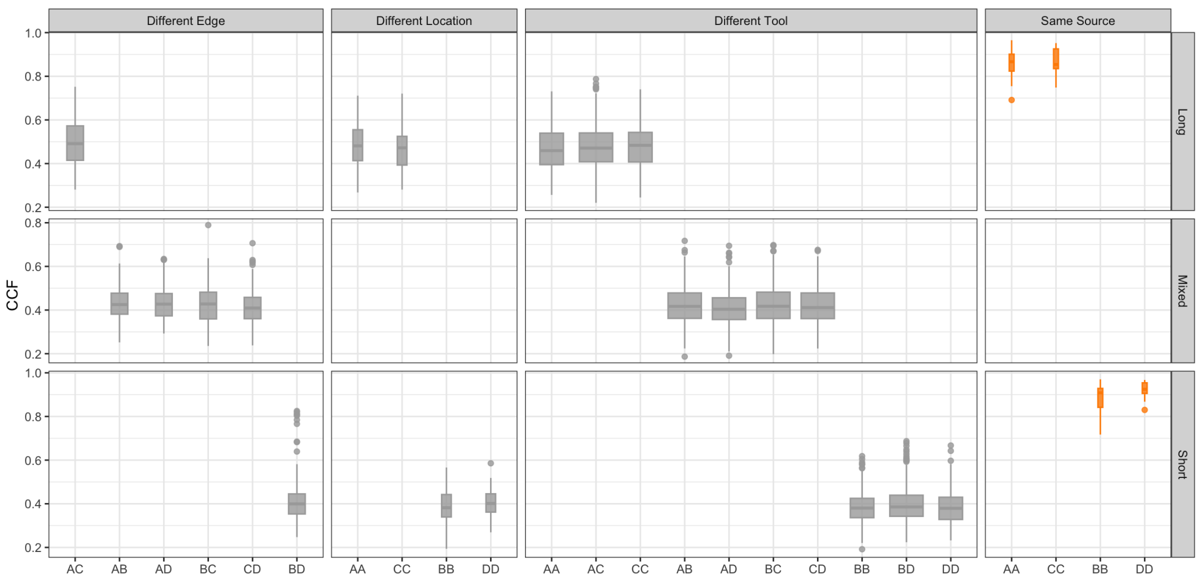

Figure 9. CCF distributions across comparison classes. Gray boxplots show different-source comparisons grouped by different edge, location, or tool; orange boxplots show same-source comparisons. Rows distinguish long, mixed, and short edge comparisons.

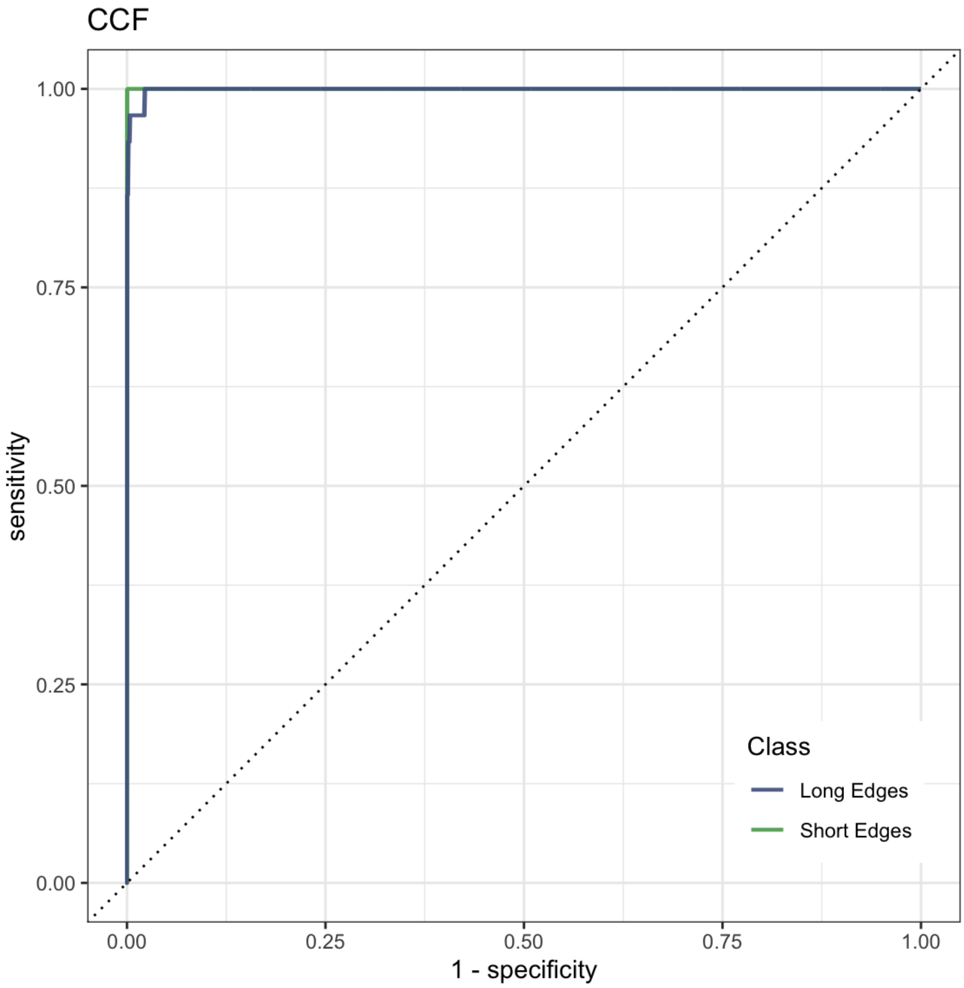

Figure 10. ROC curves for CCF-based classification. Curves are shown separately for long and short edge classes; the dotted diagonal indicates chance-level classification.

Table 2. Comparison of CCF means and standard deviations across different replicates, stagings, and acquisition settings. Dashes indicate that the standard deviation could not be calculated because only one CCF value was available.

	Replicate (R)		**Staging (S)**			**Acquisition (A)**		
Edge	**Avg**$_R$	**SD**$_R$	**Avg**$_S$	**SD**$_S$	**Avg**$_S$ **-** **Avg**$_R$	**Avg**$_A$	**SD**$_A$	**Avg**$_A$ **-** **Avg**$_S$
A	0.818	0.015	0.961	0.012	0.143	0.992	0.003	0.031
B	0.879	0.031	0.922	0.042	0.043	0.939	—	0.017
C	0.841	0.022	0.952	0.021	0.111	0.959	—	0.007
D	0.797	0.011	0.966	0.017	0.169	0.986	—	0.020

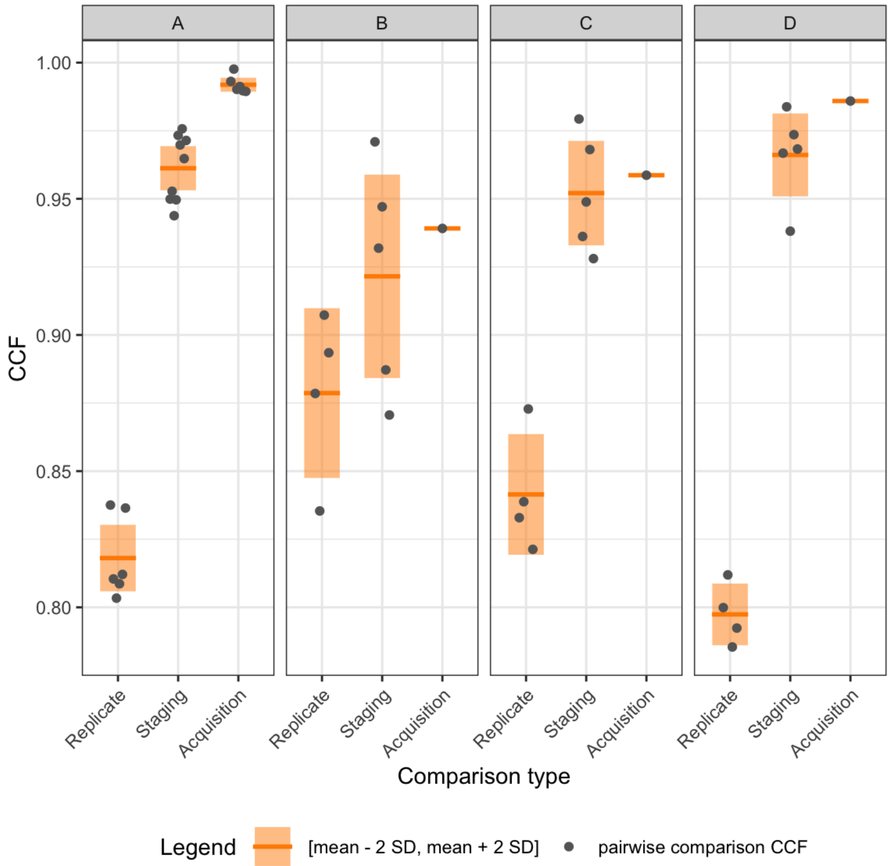

Figure 11. Variability across scanning conditions. Panels A, B, C, and D correspond to the four cut surfaces. Gray points show pairwise CCF values, orange horizontal lines show means, and shaded intervals extend two standard deviations above and below each mean.

here, we have included the extraction locations and values for each scan.

Usage Notes

The raw surface scans are provided in `x3p` format. Users who wish to inspect or process these files in R can use the `x3ptools` package and its reference manual, available from CRAN[21]. The `meta.csv` file links each scan to its tool, edge, blade location, and replicate identifiers. Users who do not wish to repeat manual profile extraction can begin with the supplied profiles and processed signals. The supplied extraction coordinates allow these derived files to be related back to the corresponding scans. Because each supplied signal is based on a manually selected profile, users assessing sensitivity to profile placement should re-extract profiles from the raw `x3p` scans using alternative documented locations.

Data Availability

The data described in this Data Descriptor are available from Iowa State University DataShare [DATASET CITATION] at `https://doi.org/[DOI]`. [Agent: Replace the dataset citation and DOI placeholders with the same record used in Data Records.]

Code Availability

We used custom R code to inspect raw scans, extract profiles, derive signals, align signals, and generate the validation outputs. The code is available in the `assessment/code` folder at `https://github.com/heike/wirecuts-data`. **[Agent: For permanent access, archive the exact code release in a DOI-providing repository and cite it if feasible; Scientific Data recommends combining GitHub with a DOI archive.]** The files and their roles are described in Table 3.

The provided code reproduces the processing, comparison, validation, and visualization outputs described in Methods and Technical Validation. **The workflow was run using R [version] and `x3ptools` [version]. Versions of additional R packages are recorded in [environment file or repository documentation path]. [Agent: Replace the R, `x3ptools`, and environment-record placeholders after checking the analysis environment.]**

Table 3. Overview of available code.

Code	Description	Section
Inspect raw scans		
1-create_pngs_from_x3p.R	obtain images of x3ps in scans/	Cutting wires
Extract profiles		
2-create_profiles_from_x3p.R	manually extract profiles from each scan	Extract profiles
3-create-single-profile-file.R	create meta profile information	Extract profiles
Derive signals		
4-create_signals_from_profiles.R	derive signals from each profile	Filtered signals
Align signals		
5-create-images.R	create images for pairwise alignment	Align signals
6-align-pairwise.R	compute pairwise alignment CCF values	Align signals
7-all-comparison-results.R	visualize comparison results	Align signals

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Author Contributions

[Agent: Confirm whether all authors approve conversion of this statement to the CRediT taxonomy before revising the contributor roles.]

Y.L.: Methodology, Software, Validation, Data Curation, Writing: Draft; H.H.: Conceptualization, Methodology, Validation, Writing: Review and Editing; C.M.: Laboratory Supervision; E.A.: Physical Specimen, Scanning; J.S.: Forensic Advice; A.C.: Funding Acquisition.

All authors reviewed the manuscript.

Competing Interests

H.H. is a technical advisor to the Association of Firearm and Toolmark Examiners (AFTE), a fellow of the American Statistical Association (ASA), and a member of the ASA Forensic Science Committee. H.H. has testified as a court witness on behalf of Judge April Neubauer in the New York State Supreme Court, Criminal Term, in New York City. **[Agent: Confirm that the declaration covers every author and replace Alicia's declaration, if any.]** The authors declare the following competing interests:

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